

Corso di Laurea in Informatica A.A. 2024-2025

Laboratorio di Sistemi Operativi

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FTP, SSH, TELNET

While working on a Linux operating system, you may need to **communicate with other devices**. For this, there are some basic utilities that you can make use of.

These utilities can help you communicate with:

- networks,
- other Linux systems
- and remote users

SSH

SSH which stands for Secure Shell, It is used to connect to a remote computer securely. Compare to Telnet, SSH is secure wherein the client / server connection is authenticated using a digital certificate and passwords are encrypted. Hence it's widely used by system administrators to control remote Linux servers.

The syntax to log into a remote Linux machine using SSH is

```
SSH username@ip-address or hostname
```

```
guru99@VirtualBox:~$ ssh guru99@68.233.250.32
guru99@68.233.250.32's password:
Last login: Mon Feb 11 04:51:24 2013 from 59.177.37.89
[guru99@cp ~]$
```

Once you are logged in, you can execute any commands that you do in your terminal

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To enable SSH on your Ubuntu system, you'll need to install the OpenSSH server. Run the following command in the terminal window:

```
sudo apt install openssh-server
```

Default Port is 22, but it can be changed.

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Type the following systemctl command:

```
$ sudo systemctl status ssh
```

Default Port is 22, but it can be changed.

```
vivek@nixcraft-power9:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enab
   Active: active (running) since Fri 2018-08-17 12:39:39 CDT; 3min 36s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 3457 (sshd)
    Tasks: 1 (limit: 22118)
   CGroup: /system.slice/ssh.service
           └─3457 /usr/sbin/sshd -D

Aug 17 12:39:39 nixcraft-power9 systemd[1]: Starting OpenBSD Secure Shell server
Aug 17 12:39:39 nixcraft-power9 sshd[3457]: Server listening on 0.0.0.0 port 22.
Aug 17 12:39:39 nixcraft-power9 sshd[3457]: Server listening on :: port 22.
Aug 17 12:39:39 nixcraft-power9 systemd[1]: Started OpenBSD Secure Shell server.
```


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SSH

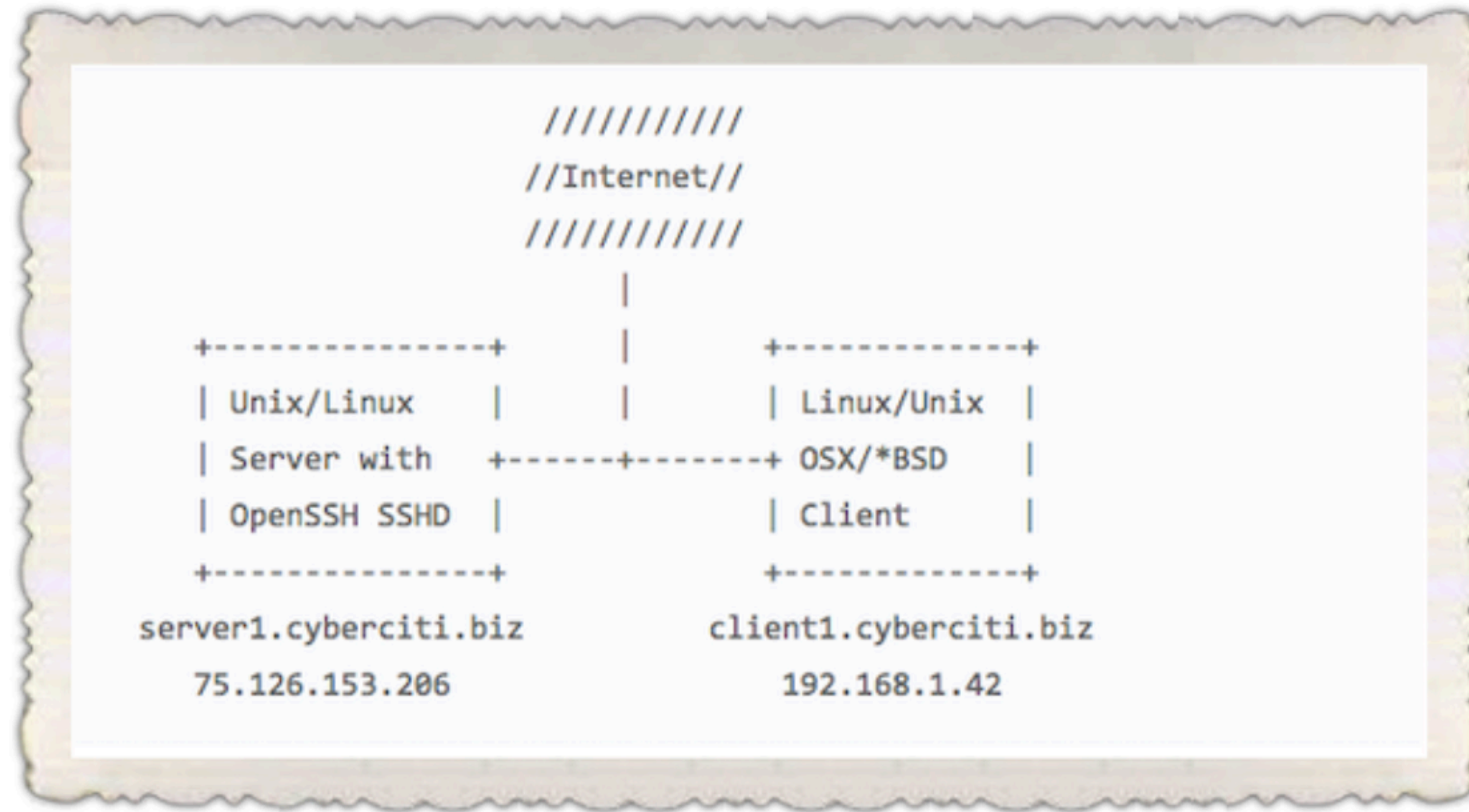
SSH server supports various authentication schema. The two most popular are as follows:

- Passwords based authentication
- Public key based authentication.

Steps to setup secure ssh keys:

1. Create the ssh key pair using `ssh-keygen` command.
2. Copy and install the public ssh key using `ssh-copy-id` command on a Linux or Unix server.
3. Add yourself to sudo or wheel group admin account.
4. Disable the password login for root account.
5. Test your password less ssh keys login using `ssh user@server-name` command.

SSH



1. `$HOME/.ssh/id_rsa` – contains your private key.
2. `$HOME/.ssh/id_rsa.pub` – contain your public key.

On the computer (such as client1.cyberciti.biz), generate a key pair for the protocol.

```
ssh-keygen -t rsa
```


SSH

The following syntax specifies the 4096 of bits in the RSA key to creation (default 2048):

```
$ ssh-keygen -t rsa -b 4096 -f ~/.ssh/vps-cloud.web-server.key  
-C "My web-server key"
```

Where,

- `-t rsa` : Specifies the type of key to create. The possible values are “rsa1” for protocol version 1 and “dsa”, “ecdsa”, “ed25519”, or “rsa” for protocol version 2.
- `-b 4096` : Specifies the number of bits in the key to create
- `-f ~/.ssh/vps-cloud.web-server.key` : Specifies the filename of the key file.
- `-C "My web-server key"` : Set a new comment.

SCP

scp - secure copy (remote file copy program)

Synopsis

```
scp [-1246BCpqrv] [-c cipher] [-F ssh_config] [-i identity_file] [-l limit] [-o ssh_option] [-P port] [-S program]
[
```

[*user@*]host1:]file1 ... [

[user@]host2:]file2

SCP

Copy file from a remote host to local host SCP example:

```
$ scp username@from_host:file.txt /local/directory/
```

Copy file from local host to a remote host SCP example:

```
$ scp file.txt username@to_host:/remote/directory/
```

Copy directory from a remote host to local host SCP example:

```
$ scp -r username@from_host:/remote/directory/ /local/directory/
```


FTP e TELNET

TELNET (TELEcommunication NETwork) and FTP (File Transfer Protocol) both are the application layer protocol. They are connection oriented protocols as they create a connection between remote host and a server.

TELNET

This software links personal computer to a network server and converts data into the plain text. Basically, it is used for remote login for a system. It makes modifications and control of the server easier.

FTP

FTP is used for transferring of web pages and used for downloading the files from other different servers. Basically, it is used for transferring the files from one system to another system more reliably and efficiently.

FTP e TELNET

TELNET	FTP
TELNET stands for TELecommunication NETwork.	FTP stands for File Transfer Protocol.
TELNET is used to access command line interface of a remote server.	FTP is used for uploading & downloading the files.
TELNET uses port 23.	FTP uses ports 20 and 21.
TELNET uses only one connection.	FTP establish two connections, one is for control command and another is for data transfer.
Remote Login is necessary in TELNET.	Remote Login is not necessary in FTP.
It does not provide high security, it provides only general security.	It provides high security than TELNET.
It is used for remote login into a system.	It is used for transferring the files from one system to another system.
It is a connection oriented protocol.	It is also a connection oriented protocol.
Telnet is mainly used for translating NVT from data and converting it to the acknowledged form.	While FTP is mainly used for transferring of data between client and server.
TELNET has become more or less outdated.	FTP is a popular tool.
TELNET is not as widely used for web diagnostics.	FTP is used for uploading and downloading web files.



Università degli Studi di Napoli Federico II

THANK YOU FOR YOUR ATTENTION

TRUST ME ...
I AM A ROBOT!

